

Acetone Gas Sensor

SMD1015 Product Datasheet



1. Product Introduction

The SMD1015 acetone gas sensor is a MEMS micro gas sensor developed based on semiconductor materials and can be used to detect the acetone gas content in exhaled breath gas.

The acetone sensing material is developed and made by IDM independently. When acetone gas reach the sensitive material, the conductivity of the material will change. A specific test circuit can be used to convert the conductance change into an output signal corresponding to the gas concentration.

2. Sensor Characteristics

It adopts MEMS technology and semiconductor materials, and has the advantages of small size, low power consumption, high sensitivity, fast response recovery, simple driving circuit, good stability, long life, solid structure and good shock resistance.

3. Main Application

It can be applied to the detection of acetone content in human exhaled breath gas especially for rapid disease detection.

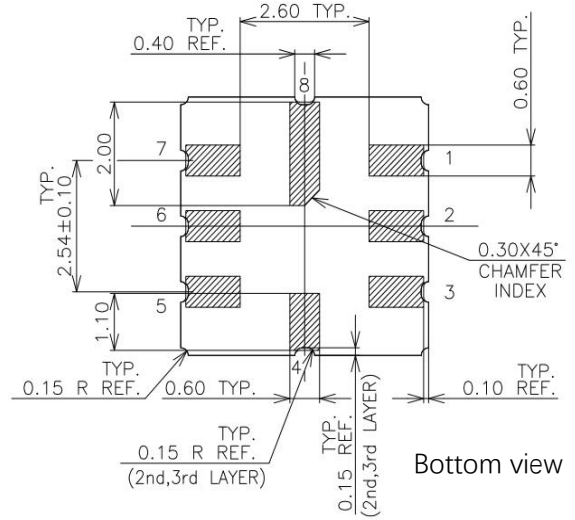
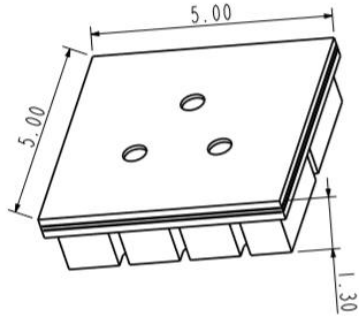
4. Product Description

4.1 Technical parameters

Table 1

Product Number			SMD1015	
Product Type			MEMS Semiconductor Sensors	
Standard Package			Ceramic Package	
Target Gas			Acetone	
Detection Concentration Range			0-500 ppm (acetone)	
Resolution			0.1 ppm (acetone)	
Working Temperature			-20 ~ 55 °C	
Working Humidity			10 ~ 90 %RH	
Standard Circuit Conditions	Sensor Voltage		V _C	5 V or 3.3 V DC
	Heating Voltage		V _H	1.8 ±0.1 V DC
	Load Resistance		R	Adjustable (subject to shipping sheet)
Gas Sensor Characteristics Under Standard Test Conditions	Heater Resistance		R _H	45 ± 5 Ω (measured at room temperature)
	Heating Power		P _H	≤30 mW
	Sensor Resistance		R ₀	10 ~ 1000 KΩ (standard condition tested in air)
	Sensitivity		S	R 0 (in air)/Rs (in 1ppm acetone) ≥ 1.5
	Response Ratio		α	≤ 0.5 (R 3 ppm / R 1 ppm acetone)
Standard Test Conditions		Temperature		20 °C ±2°C
		Humidity		55%±5%RH
		Preheat Time		3-5 min
Response Time (T ₉₀)			<30 s	
Recovery Time (T ₁₀)			<60 s	
Life Time			≥ 3 years	

4.2 Sensor structure and pin definition

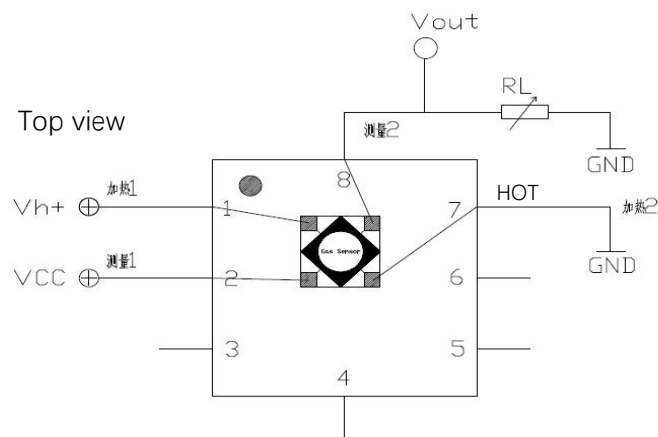


Bottom view

Table 2

Pin Number	Name	Definition
1	VH	Heating voltage supply
2	VCC	Sensor voltage supply
3	NG	/
4	NG	/
5	NG	/
6	NG	/
7	HOT	Heater GND
8	VOUT	Sensor output voltage

4.3 Basic Circuit



The figure shows the basic test circuit of the SMD1015 sensor. The sensor needs apply two voltages: heater voltage and sensor voltage. V_h is a specific heating voltage for the sensor heater. V_{out} is the output test voltage, which also connect a load resistor (R_L) in series. V_{cc} is sensor voltage.

Note: in following section, if the unit is resistance, it is calculated by this formula:

$$R = \frac{R_L(V_{CC} - V_{OUT})}{V_{OUT}}$$

Each item definition refer table1

5. Sensor Characterization

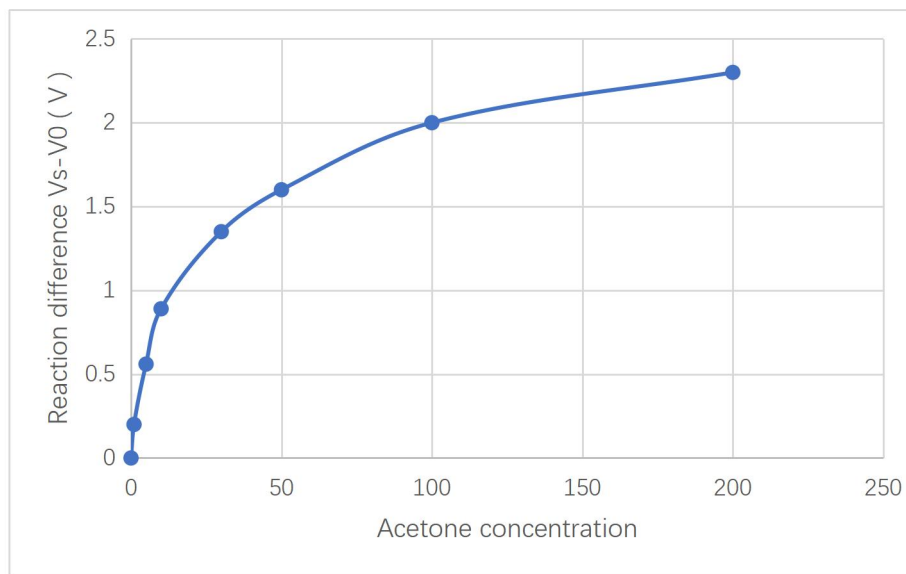


Figure 1 Sensor sensitivity characteristic curve

All tests were completed under standard test conditions, and the load resistor used was $10K\Omega$. In the figure, V_0 represents the sensor response voltage value in clean air, and V_s represents the sensor response voltage in different concentrations of target gas.

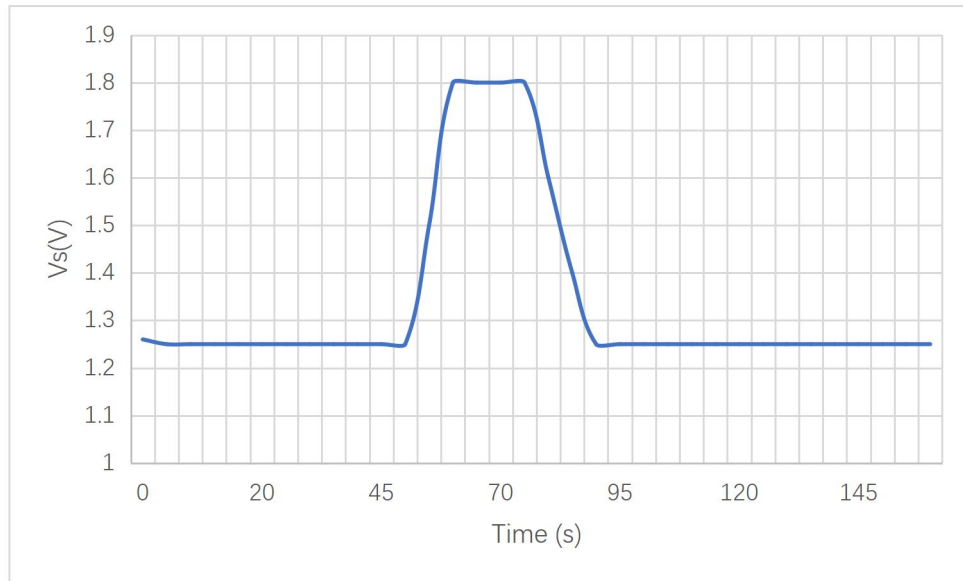


Figure 2 Sensor response-recovery characteristic curve

All tests were completed under standard test conditions, with a load resistance of 10KΩ. In the figure, V_s represents the response voltage of the sensor, and the test gas is 30ppm acetone.

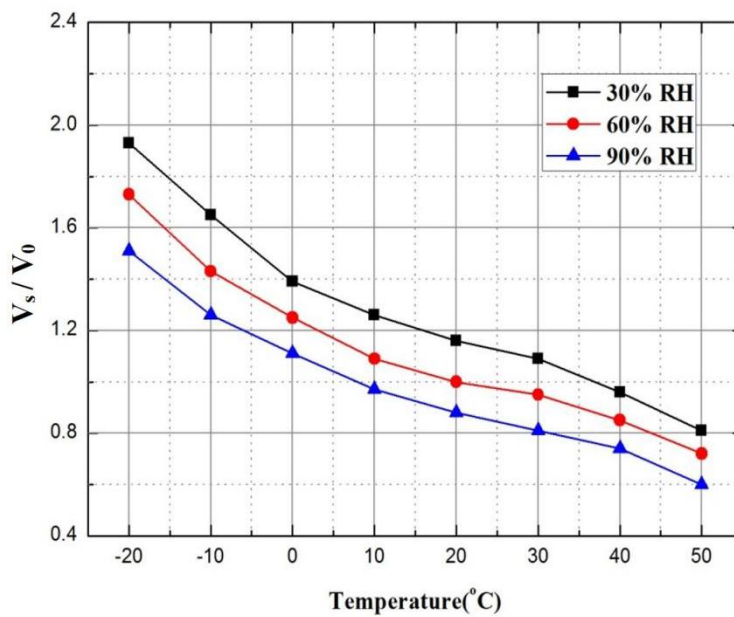


Figure 3 Sensor temperature and humidity characteristic curve

Above figure shows the temperature and humidity effect curves, V_s represents the response voltage in 30 ppm acetone gas under different temperature and humidity conditions, and V_0 represents the response voltage value in clean air under correspond temperature and humidity conditions.

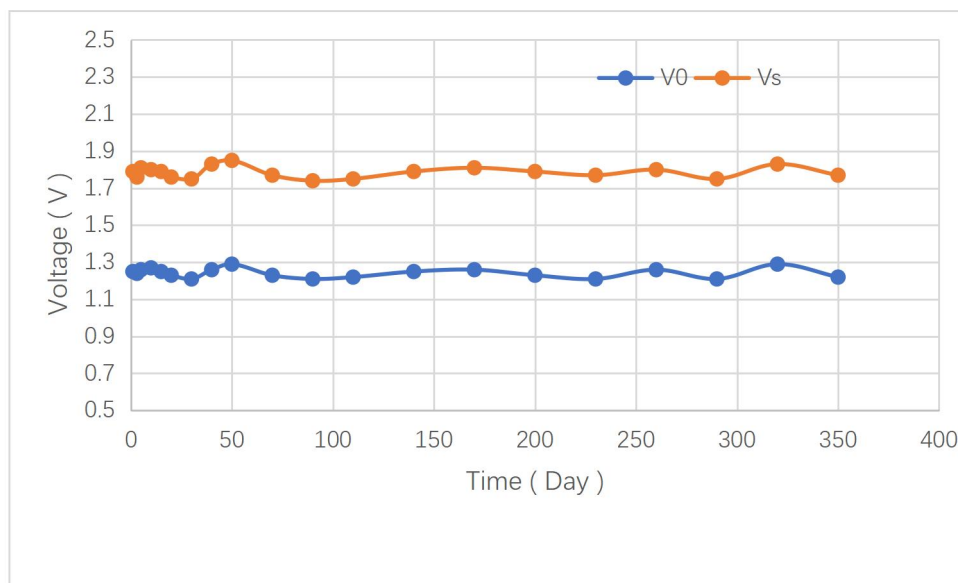


Figure 4 Long-term stability curve of the sensor

All tests in the figure were completed under standard test conditions, and the load resistance was 10KΩ. V0 represents the sensor response voltage in clean air, Vs represents the sensor response voltage in a 30ppm acetone, and the horizontal axis represents the test time when sensor is continuous powered on.

6. Product Shipping Packaging:

Generally shipped in carrier tape packaging. Customer can request other package when place the purchase.

Precautions:

1. Situations must avoid

1.1 Exposure to volatile silicon compound vapor

The sensor should avoid being exposed to silicone adhesives, hair spray, silicone rubber, putty or other places where volatile silicone compounds exist. If the surface of the sensor is adsorbed with silicone compound vapor, the sensitive material of the sensor will be wrapped by silicon dioxide formed by the decomposition of the silicone compound, which will degrade the sensitivity of the sensor and cannot be restored.

1.2 Highly corrosive environment

The sensor is exposed to high concentration of corrosive gas (such as H₂S, SO₂, Cl₂, HCl, etc.), it will not only cause corrosion or damage to the heating material and sensor leads, but may also cause irreversible deterioration of the performance of sensing materials.

1.3 Pollution by alkali, alkali metal salts and halogens

The performance of the sensor may also deteriorate when it is contaminated by alkali metals, especially salt water spray, or exposed to halogens such as Freon.

1.4 Directly touch with water

Splashing or immersing the sensor in water will cause the sensor's sensitivity to decrease.

1.5 Freeze

If sensor surface appear with ice, it will cause sensing material crack and lost sensitive property.

1.6 Applied voltage is too high

If the voltage applied to the sensor or heater is higher than the specified value, even if the sensor is not physically damaged or destroyed, it may cause damage to the leads and/or heater and cause the sensor's sensitivity characteristics to degrade.

1.7 Voltage pins connect mismatch

If the wrong voltage is applied to the sensor or heating and signal pins, it may cause damage to the leads and/or heater and cause the sensor's sensitivity to deteriorate.

2. Situations need attention

2.1 Condensation

Under indoor use conditions, slight condensation will have a slight effect on sensor performance. However, if water condenses on the surface of the sensitive layer and remains for a long time, the sensor characteristics will deteriorate.

2.2 In high concentration gas

Regardless of whether the sensor is powered on or not, long-term exposure to high-concentration gas will affect the sensor's characteristics. For example, spraying lighter gas directly at the sensor will cause great damage to the sensor.

2.3 Long-term storage

If the sensor is stored for a long time without power, its resistance will have a reversible drift, which is related to the storage environment. The sensor should be stored in a sealed bag that does not contain volatile silicon compounds. After long-term storage, the sensor needs to be powered for a longer time before use to stabilize it. The storage time and corresponding aging time are recommended as follows:

Storage time	Recommended preheating time
Less than 1 month	Not less than 6 hours
1-6 months	Not less than 12 hours
More than 6 months	No less than 24 hours

2.4 Long-term exposure to extreme environments

Regardless of whether the sensor is powered or not, prolonged exposure to extreme conditions such as high humidity, high temperature or high pollution will seriously affect sensor performance.

2.5 Vibration

Frequent, excessive vibration can cause the sensor's internal leads to resonate and break. This type of vibration can be generated during transportation and by using pneumatic screwdrivers/ultrasonic welders on the assembly line.

2.6 Excessive impact

If the sensor is subjected to a strong impact or falls, its lead wires may break.

2.7 Conditions of assembling:

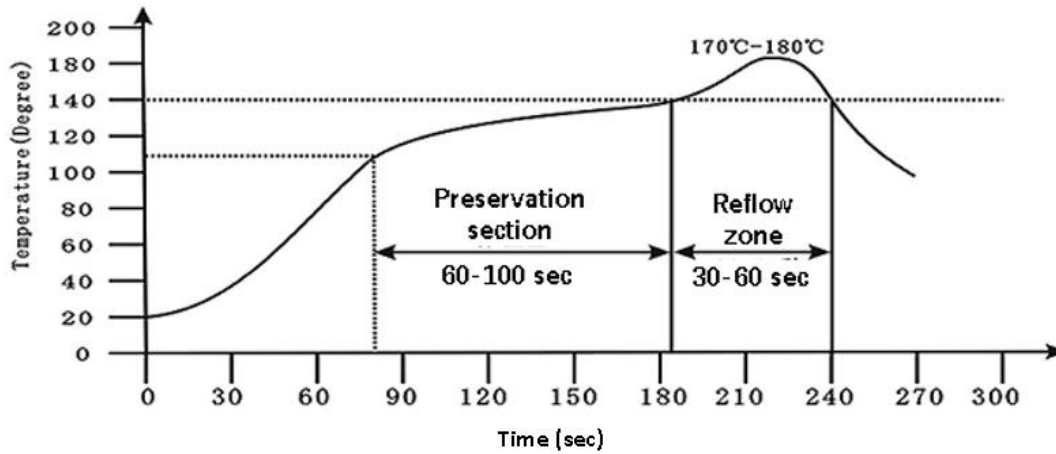
2.7.1 Manual welding is the most ideal welding method for sensors. The recommended welding conditions are as following:

- Flux: Rosin flux with minimal chlorine content
- Constant temperature soldering iron
- Temperature: 250℃
- Time: no more than 3 seconds

2.7.2 The following conditions are recommended when using reflow soldering:

Solder paste: low temperature lead-free solder paste (Sn42Bi58)

The furnace curve is as following:



2.8 Anti-static

Anti-static bag packaging

Violation of the above usage conditions will degrade the sensor characteristics.

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